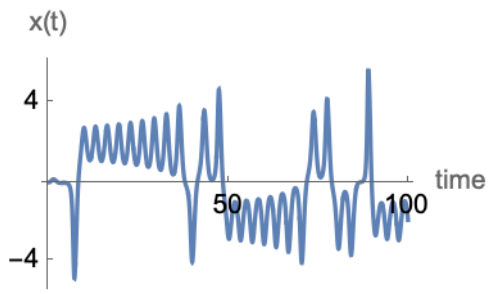
Class 24 Introduction to chaotic dynamics

Questions

Teams 5 and 6: Post screenshots of your work to the course Google Drive today. Include words, labels, and other short notes that might make those solutions useful to you or your classmates. Find the link in Canvas.

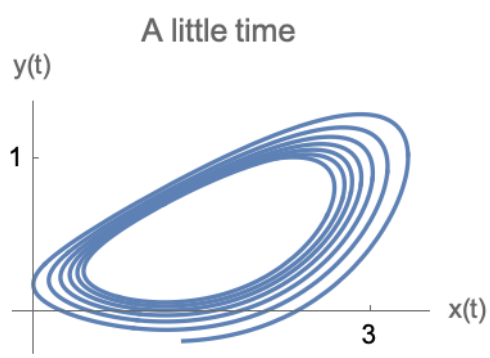
1. The following plot shows $x(t)$ vs t for a numerical solution of a 3D dynamical system.

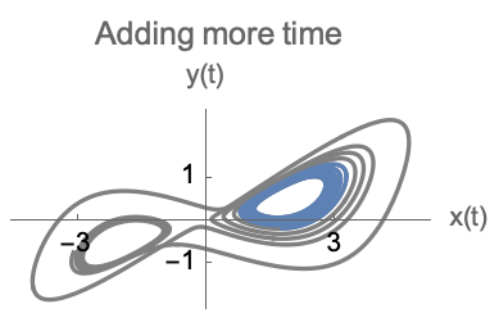
   - What features do you notice in this plot of the single variable versus time?
   - Does the motion appear periodic, nearly periodic, or irregular? Describe what you see.
   - If you only had this plot, what would you guess about the long-term behavior?



2. The following plot shows $(x(t), y(t))$ plotted in the xy-plane for the same numerical solution as above. This is a solution to a 3D dynamical system, so this xy view is a projection.

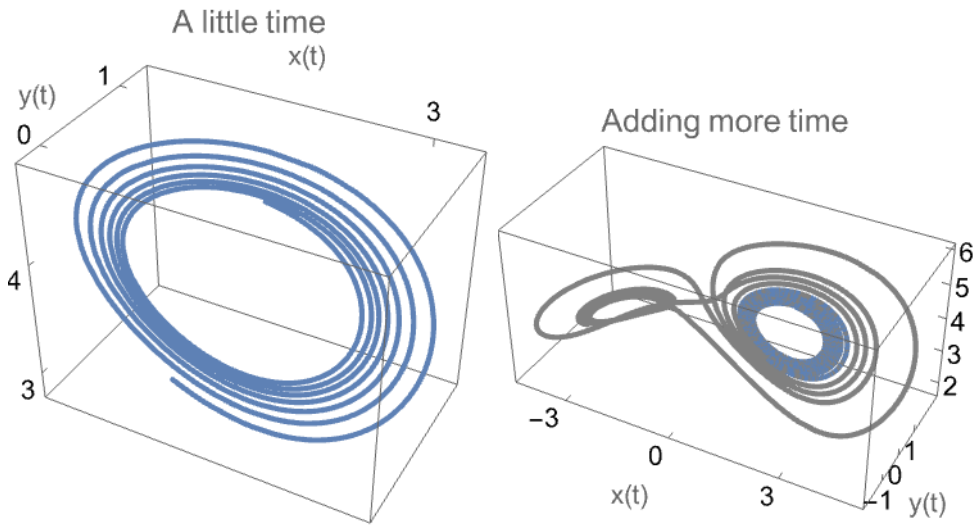
   - Describe the overall shape of the trajectory in each 2D view.
   - Does the trajectory appear to cross itself in these plots?
   - If it appears to cross itself, does that mean the actual solution in state-space crosses itself? Why or why not?
   - What do these plots suggest about the long-term behavior of the system?





3. Now we look at all three variables.

   - How does the full 3D trajectory compare with what you observed for the 1D and 2D plots?
   - In the full 3D phase space, does the trajectory cross itself?
   - What seems to be the long-term behavior: approaching an equilibrium, a periodic orbit, or something more complicated?



3D systems: Example 1

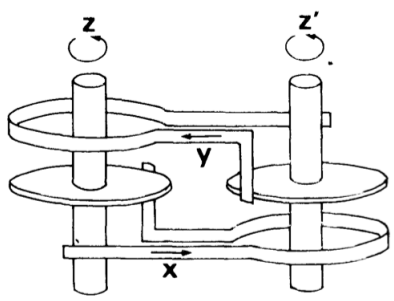
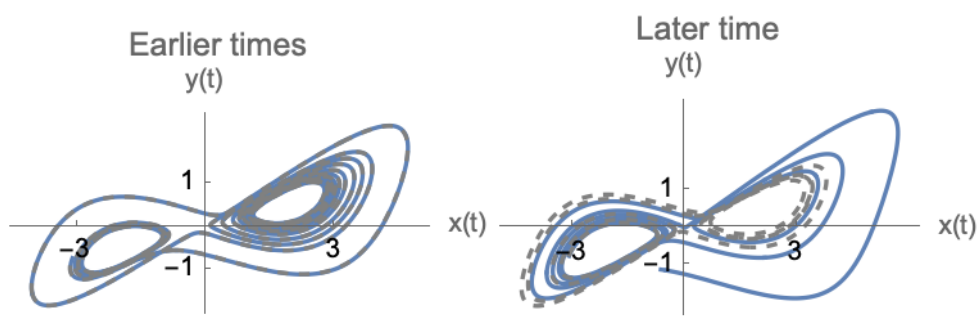


Fig. 1. Rikitake two-disc dynamo system.

Figure from (Ito 1980).

4. Next we look at two trajectories that start at slightly different initial conditions.



- At early times, how similar are the two trajectories?
- What happens at later time? Do they continue to stay together or do they separate?
- Do they seem to remain in the same overall region of phase space?
- Describe how the distance $d(t)$ between the two trajectories changes over time.
  Do you think $d(t)$ vs t should show
  - convergence to 0?
  - staying about the same distance apart over time?
  - divergence?

5. The plot below shows the $\log_{10}$ of the distance between the trajectories.

(a) How does this graph compare with your prediction?

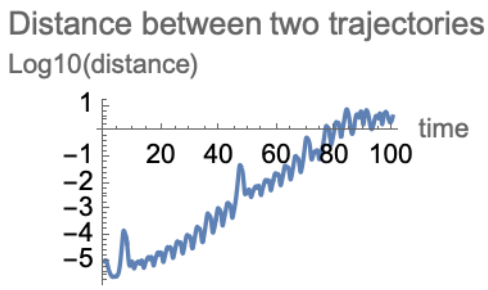
(b) Over what time interval is $\log d(t)$ vs t (very approximately) a straight line?

(c) If $\log d(t) \approx mt + b$ on some interval, how is $d(t)$ growing?

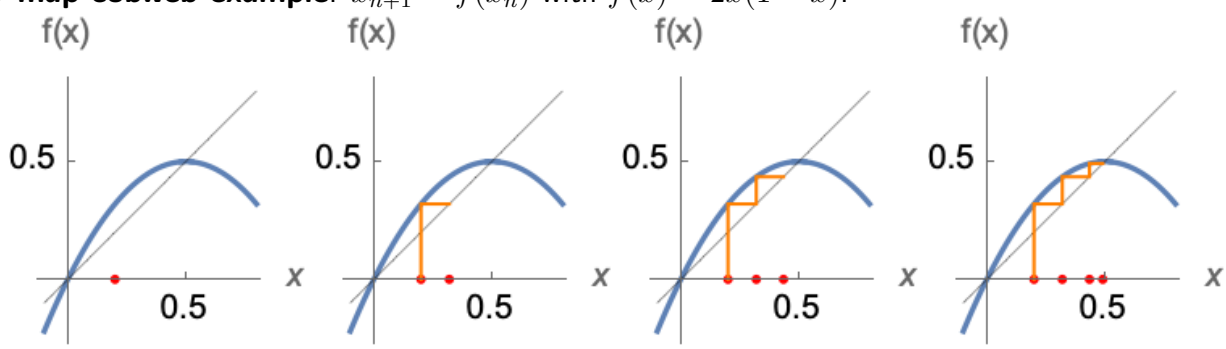
(d) What does the slope m indicate about how quickly nearby trajectories separate?

(e) The graph eventually stops growing linearly and levels off. Comparing to what you see in the trajectories, why do you think that happens?

(f) Use this example to say what "sensitive dependence on initial conditions" might mean.



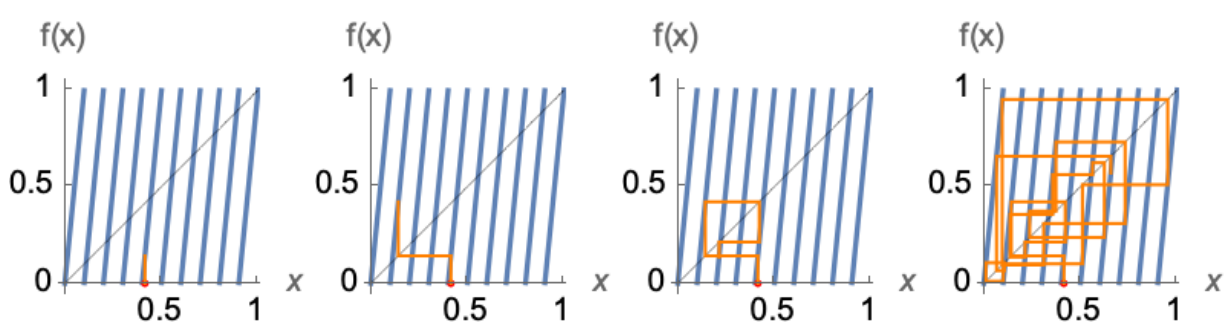
1D map cobweb example. $x_{n+1} = f(x_n)$ with $f(x) = 2x(1-x)$.

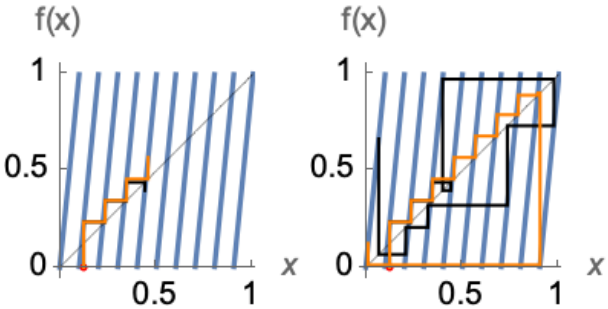


Practice.

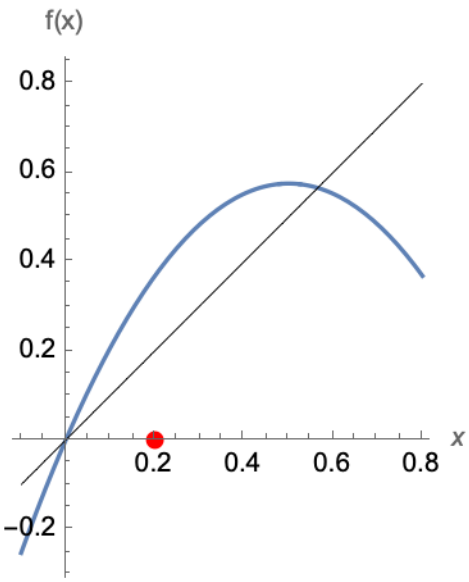
Let $x_{n+1} = f(x_n)$ with $f(x) = \cos x$.

- Plot $\cos x$ and x on the same plot.
- Use your plot to identify any fixed points of the map.
- Make cobwebs for at least two qualitatively different cases.





6. Let $x_{n+1} = f(x_n)$ where $f(x) = 2.3x(1-x)$. Add a cobweb to the graph to represent the orbit, using the red dot as your starting value of x.



7. (9.4.2) The tent map is a simple map Let

$$x_{n+1} = \begin{cases} 2x_n, & 0 \le x_n \le \frac{1}{2} \\ 2 - 2x_n, & \frac{1}{2} \le x_n \le 1. \end{cases}$$

   (a) Draw $f(x)$ where $x_{n+1} = f(x_n)$. How many times does it intersect the curve $y = x$? Why is this map the "tent map"?

   (b) Find the fixed points of this map.
   This is a 1d map, so the location of a fixed point is given by a single number.

   (c) Classify the stability of the fixed points. *Use cobwebbing or compare the slope of $f(x)$ at the fixed point to 1.*